AERIS-10 - end-to-end signal and control data flow (antenna -> RF -> ADC -> FPGA -> host; host -> STM32 -> clock/LO/beamformer -> FPGA)
Drawing ID: SD-07 Revision: A Date: 2026-10-09 Status: PARTIAL - the FPGA-to-host data path has NO hardware (FT601 unwired) and several interfaces are unverified
Source: docs/SYSTEM/BLOCK_DIAGRAM.md (EAGLE connector nets + firmware + RTL), docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md 6-7, docs/STM32/STM32_PROJECT_RECONSTRUCTION.md 3.2-3.3 and STM-T04, docs/GUI/DEPENDENCIES.md 4 and 6,
9_Firmware/9_2_FPGA/reconstructed/PIN_MAP_FROM_SCHEMATIC.md, 9_Firmware/9_2_FPGA/radar_system_top.v, 9_Firmware/9_1_Microcontroller/9_1_3_C_Cpp_Code/main.cpp, 9_Firmware/9_3_GUI/GUI_V5.py
Edge styles: SOLID = confirmed by schematic net AND firmware/RTL reference; DASHED = interface exists in one source only or mapping unverified; DOTTED RED = missing specification / no hardware path

LEGEND
SOLID black edge = interface confirmed by a schematic net AND a firmware/RTL reference (file:line in label)
DASHED edge = unverified: present in only one source, anonymous net, inferred mapping, or wiring confirmed but driver/RTL broken (orange = documented defect)
DOTTED RED edge = missing specification or no hardware path at all
red dotted box = component with no wiring/CAD; orange box = documented conflict (K-numbers from docs/SYSTEM/BLOCK_DIAGRAM.md section 4; C-numbers from docs/STM32/STM32_PROJECT_RECONSTRUCTION.md STM-T04)
KEY FINDING: the only wired host link is STM32 USB-FS CDC (control + GPS/status). The FPGA radar-data output (usb_data_interface.v) targets an FT601 that has no nets;
no GUI version can decode the RTL packet format (K5). Therefore no end-to-end radar data path exists in the repository as committed.
NOT VERIFIED: any electrical measurement, RF levels, LO/IF frequency plan beyond the constants quoted, cable harness, antenna

RF FRONT END
Antenna array
README: 8x16 patch or 32x16 slotted waveguide
NO CAD, NO PART, two different simulations (BLOCK_DIAGRAM 1, K8)
RF (antenna interface unspecified)
17x M3SWA2-34DR+ switches
16+1 SMA pairs J22..J55 to PA boards
17th pair UNRESOLVED (BLOCK_DIAGRAM 2.3)
schematic
16x ADTR1107 (T/R front end)
schematic
4x ADAR1000 beamformers
1.8 V SPI/TR/LOAD from FPGA bank 34
(ADAR1000_Manager.cpp via hspi1 -> FPGA pass-through)
beamformed IF/RF (schematic)
2x LTC5552 mixers U5, U13
TX up-conversion / RX down-conversion
SMA pairs J22..J55 (17 pairs, 16 PAs; mapping of the 17th unresolved)
RF PA boards x16 (AERIS-10X)
QPA2962 GaN 10 W, 35x60 mm 4L
J1 RFIN / J2 RFOUT; X3 VD,VIN_M; X2 VG
RX IF 120 MHz into AD9484 (schematic; DDC IF_FREQ=120e6 ddc_400m.v:48)
MAIN BOARD 260x300 mm 10L (digital / mixed-signal)
AD9484 ADC U1, 8 bit 400 MSPS
clock 400 MHz AD9523 OUT4 (J3 -> J21)
LVDS D0..7 + DCO to FPGA bank 14
LVDS D0..7 + DCO, bank 14 at 3.3 V (schematic) vs RTL LVDS_25 + DIFF_TERM (ad9484_interface_400m.v:25-27); capture module ad9484_lvds_to_cmos_400m MISSING - PIN_MAP:9-10
FPGA XC7A50T-2FTG256I U42 (CAD) / XC7A100T (README+XDC) - conflict K1
radar_system_top: TX chirp -> DAC; ADC capture -> DDC -> matched filter -> Doppler -> packetizer
RTL status: 5 missing modules, LVDS capture not implemented (see SD-01/SD-02)
usb_data_interface.v FT601 slave FIFO: NO NETS on the board
FT601 USB 3.0 FIFO U6
PLACED BUT 0 OF 77 PINS CONNECTED
(PIN_MAP_FROM_SCHEMATIC.md:77; BLOCK_DIAGRAM 1)
radar data (range/Doppler/detections): NO HARDWARE PATH EXISTS - GUI expects FT2232H/FT601 data (GUI_V5.py:479-549)
USB 3.0: no connector, no cable, no driver in GUI (pyftdi cannot drive FT60x)
Host PC - Python GUI
GUI_V5.py: STM32 CDC (pyusb) + FTDI FT2232H (pyftdi) - no FT2232H exists on the board
GUI_V6_Demo.py: offline simulator
radar packet format A5C3+CRC16 (GUI) vs 0xAA..0x55 (RTL) - conflict K5
USB CDC: start flag [23,46,158,237] + 82-B settings (GUI_V5.py:401-468); status text / GPSB back (main.cpp:1692, gps_handler.cpp:118) wiring confirmed; firmware RX callback never bound (C4) - see SD-04
handshake DIG_0..4 = PD8 new_chirp, PD9 new_elevation, PD10 new_azimuth, PD11 mixers_enable, PD12 reset_n (schematic nets DIG_0..4 -> F13/E16/D16/F15/E15; main.cpp:448,484,514,1483,1660; radar_system_top.v:23,71-74) bit-to-port mapping inferred from source comments (MEDIUM, PIN_MAP:16-20)
SPI1 PA5/PA6/PA7 + CS PA0..PA3 -> FPGA J16/G14/H13, F14/H16/G16/H15 (schematic; ADAR1000_Manager.cpp:659, main.h:78-85) RTL pass-through stm32_*_1v8 UNDRIVEN (radar_transmitter.v:65-71)
DIG_5..7 (PD13..PD15 inputs, never read; no RTL port) -> H11/G12/H12
X53 mini-USB
STM32_USB_FS_D_P/D_N/ID (PA11/PA12/PA10)
the ONLY host data link in the CAD (BLOCK_DIAGRAM 2.4)
STM32_USB_FS_D_P/D_N (schematic) <-> USB OTG_FS (usbd_conf.c missing)
DAC_0..7 bank 15 (schematic) = dac_data[7:0] radar_system_top.v:28 (dac_clk port has no board pin, PIN_MAP:8)
STM32F746ZGT7 U2
main.cpp: power sequencing, AD9523/ADF4382 config (SPI4), ADAR1000 (SPI1), PA bias (I2C1/I2C2), sensors (I2C3/UART5), stepper (PD4/PD5), FPGA handshake (PD8..PD12), USB CDC to host
I2C1 PB6/PB7, I2C2 PF0/PF1 (schematic + main.cpp:1563-1657)
PA bias: 2x DAC5578 (I2C1 @0x48/0x49) -> 16x VG; 16x INA241A3 -> 2x ADS7830 (I2C2 @0x48/0x4A) Idq
8x TMP37 -> ADS7830 @0x49
(main.cpp:1563-1657, 1670-1753)
VG / VD / VIN_M via X3,X38..X52 3-pin Molex (anonymous nets)
EN_* x16 via SV1<->SV1 (schematic + main.h:94-128, main.cpp:1241-1275,1485-1489)
rails via 2-pin Molex (schematic 2.1); several rail names differ or are not found - verify
POWER SUPPLY BOARD 280x300 mm 2L
X1 VIN 12-17 V -> 34 rail outputs
X2..X35; EN bus SV1 (16x EN_*)
PA 22 V requirement (xlsx) has no rail - conflict K4
+5V0_PA_1..3, +5V5_PA (schematic); 22 V PA drain (xlsx) has no rail
AD9708 DAC U3, 8 bit
DAC_0..7 from FPGA bank 15; clock 120 MHz AD9523 OUT10 via J20
1.8 V SPI + TR + LOAD lines, bank 34 (schematic; RTL ports radar_system_top.v:40-57) SPI never driven in RTL; TR/LOAD from plfm_chirp_controller
LO TX SMA (anonymous nets J1/J2/J5/J6/J9-J13) - mapping requires schematic review
LO RX SMA (anonymous nets) - mapping requires schematic review
SPI4 + PG11..PG15 via JP13<->JP2 (schematic + main.h:144-157); firmware driver broken: platform_ops=NULL (C3), pin macros collide (C2)
SPI4 + PG6..PG10 via JP13<->JP2 (schematic + main.h:124-143); same firmware defects C2/C3
UART5 PC12/PD2, I2C3 PA8/PC9, GPIO PD4..PD7 (schematic connectors + main.cpp:908, 1280, 1514-1521)
module part numbers and pinouts unspecified
SPI4 PE2/PE5/PE6 + PF3..PF10 control via JP1<->JP1 (schematic + main.cpp:927-1058, stm32f7xx_hal_msp.c:323-324)
FREQUENCY SYNTHESIZER BOARD 100x100 mm 6L
OCXO 100 MHz X4 / VCXO 50 MHz X5,X6 (schematic) vs xlsx VCXO 100 MHz / firmware vcxo_freq 100 MHz - conflict K7
ADF4382 U1 - TX LO (10.5 GHz per adf4382a_manager.c:54)
ADF4382 U6 - RX LO
Off-board modules (no part numbers): GPS (JP8, UART5 9600), GY-85 IMU + BMP180 (JP7/JP18, I2C3), stepper driver (JP9, PD4/PD5), fan relay (JP4, PD7), PA drain switch (JP10, PD6)
+5V0_LO, +3V3_LO_1/2, +1V8_CLOCK, +3V3_CLOCK, +3V3_XO (anonymous synth connector nets)
reference (schematic); oscillator set conflicts with firmware (K7)
AD9523 clock generator IC1
SPI4 + control via JP1<->JP1; CS PF7 never toggled (C7)
OUT4 400 MHz ADC, OUT5 400 MHz FPGA (unused by RTL), OUT6 100 MHz FPGA sys, OUT7 20 MHz test, OUT10/11 120 MHz DAC/FPGA
OUT10 120 MHz -> J20 DAC_CLOCK (anonymous synth net; firmware main.cpp:1025-1026)
OUT6 100 MHz J7 -> J1 FPGA_SYS_CLOCK (schematic + main.cpp:995-996 + RTL clk_100m)
OUT5 400 MHz J19 FPGA_ADC_CLOCK (schematic; unused by RTL)
OUT11 120 MHz -> J18 FPGA_DAC_CLOCK = RTL clk_120m_dac (anonymous synth net)
OUT4 400 MHz J3 -> J21 (schematic + main.cpp:983-984)
TX chirp (schematic)